

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Draw the pin diagram of 8051 microcontroller. Explain any four types of interrupts used in 8051 microcontroller.

Q.24 Describe the function of the SJMP instruction in the context of program control in the 8051 microcontroller. What does the a CALL instruction do in the 8051 instruction set?

Q.25 Discuss the trends and advantages in embedded system technology, including IoT (Internet of Things) integration and edge computing.

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Roll No.

4th Sem / Instrumentation & Control

Subject : Microcontroller and Embedded Systems

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 The internal RAM memory of 8051 is

- a) 32 bytes b) 64 bytes
- c) 128 bytes d) 256 bytes

Q.2 Which of the following is the correct addressing mode used by the 8051 for accessing data from external memory?

- a) Direct addressing mode
- b) Register indirect addressing mode
- c) Indexed addressing mode
- d) Immediate addressing mode

Q.3 Which of the following is NOT a typical component of an embedded system?

- a) Microcontroller b) Operating system
- c) Sensors d) Display monitor

Q.4 The INC instruction in the 8051 is used for :

- a) Incrementing a register
- b) Decrementing a register
- c) Incrementing the accumulator
- d) Decrementing the accumulator

Q.5 What is the size of the program memory (ROM) in the standard 8051 microcontroller?

- a) 2 kilobytes b) 4 kilobytes
- c) 8 kilobytes d) 16 kilobytes

Q.6 How many Ports 8051 is having

- a) 1 b) 2
- c) 3 d) 4

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Expand RISC.

Q.8 In the 8051 microcontroller, _____ register is used for serial communication?

Q.9 'MOV A, 40H' moves the value stored in the memory location 40H into the accumulator A. (True/False)

Q.10 Expand AVR.

Q.11 8051 has a _____ bit data bus.

Q.12 Define microcontroller.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Write the major differences between microprocessor and microcontroller.

Q.14 Explain all register which are used in 8051 microcontroller.

Q.15 Draw and explain program status word.

Q.16 Explain the function of the ADD and INC instruction in the 8051 microcontroller.

Q.17 What is the significance of the JMP instruction in the 8051 instruction set?

Q.18 Explain the role of microcontrollers in embedded systems and provide examples of popular microcontroller families.

Q.19 Describe the typical components of an embedded system and their functions.

Q.20 Explain the key characteristics of the ARM microcontroller architecture.

Q.21 Describe the primary architectural features of AVR microcontrollers.

Q.22 Write down the applications of microcontroller.